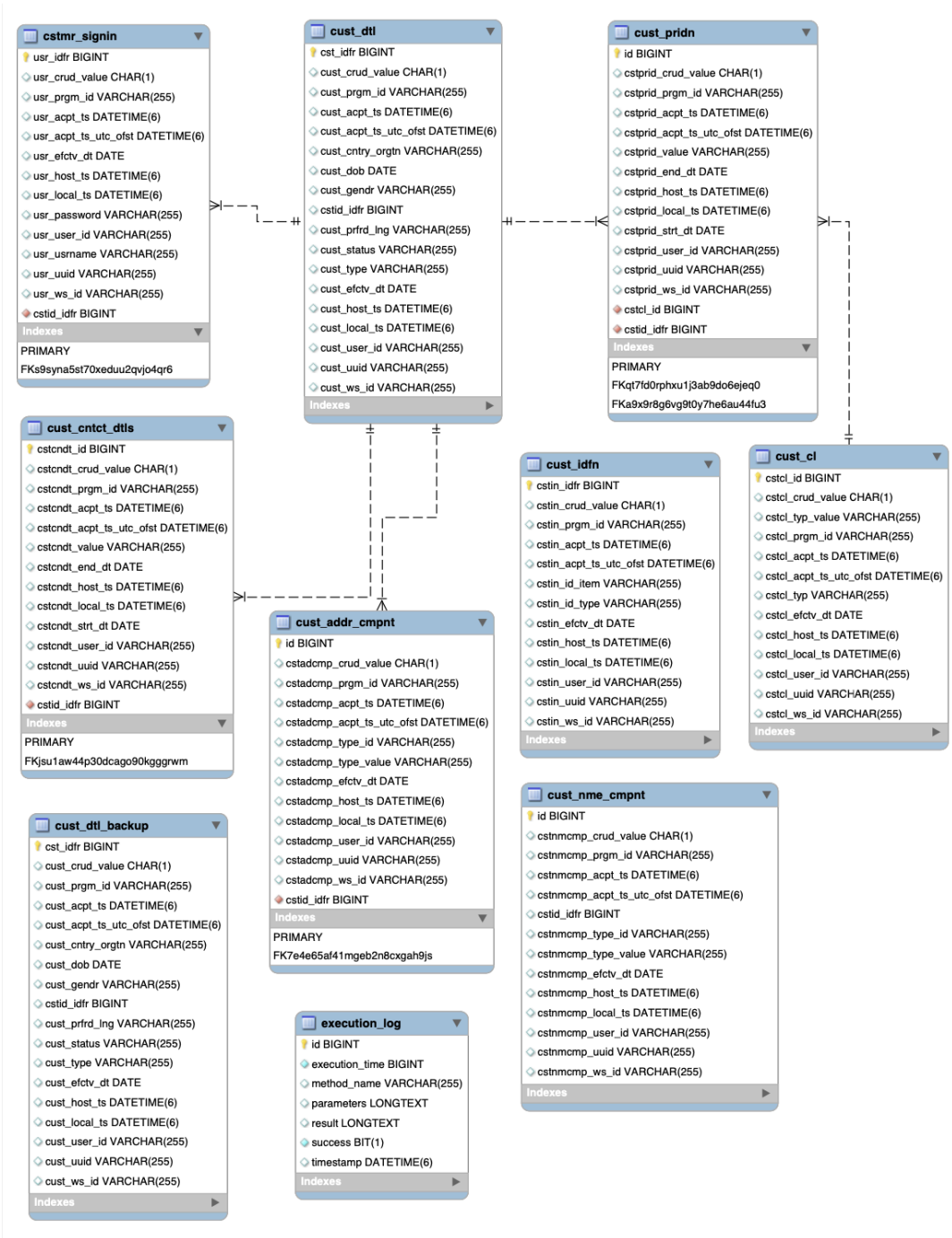
FinTech Lab
Week 2

Name: Sharanya S Yelluru
Registration No: 2409658176
Roll no: 42
Section: CSFT A
Date:15th-16th January 2026

Lab manual questions-

- ER diagrams:



- Tables in database:

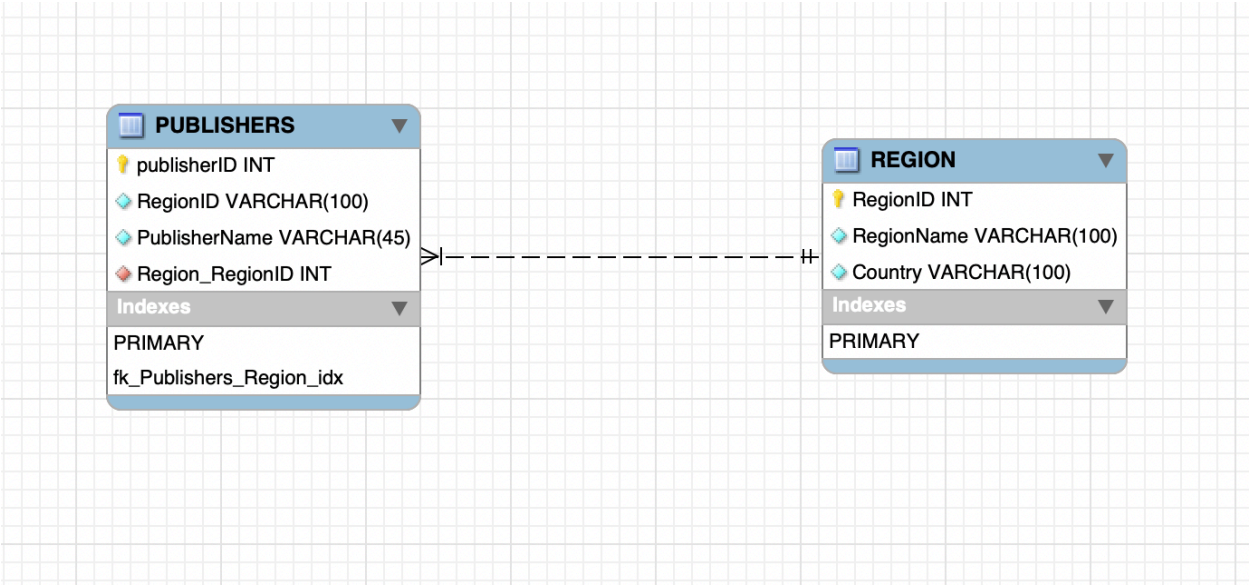
```
mysql> show databases;
+--------------------+
| Database           |
+--------------------+
| information_schema |
| lab1               |
| LAB2               |
| LAB3               |
| lab5               |
| mit                |
| mysql              |
| performance_schema |
| sys                |
+--------------------+
9 rows in set (0.003 sec)

mysql> use mit;
Database changed
mysql> show tables;
+-----------------+
| Tables_in_mit   |
+-----------------+
| cstmr_signin    |
| cust_addr_cmpnt |
| cust_cl         |
| cust_cntct_dtls |
| cust_dtl        |
| cust_dtl_backup |
| cust_idfn       |
| cust_nme_cmpnt  |
| cust_pridn      |
| execution_log   |
+-----------------+
10 rows in set (0.004 sec)
```

PPT ER Diagram Questions:

1. A publishing ecosystem stores details of **Regions** and **Publishers**. Each **Publisher** operates in exactly one **Region**, but a **Region** can have many **Publishers**. Draw the ER diagram and show cardinalities.

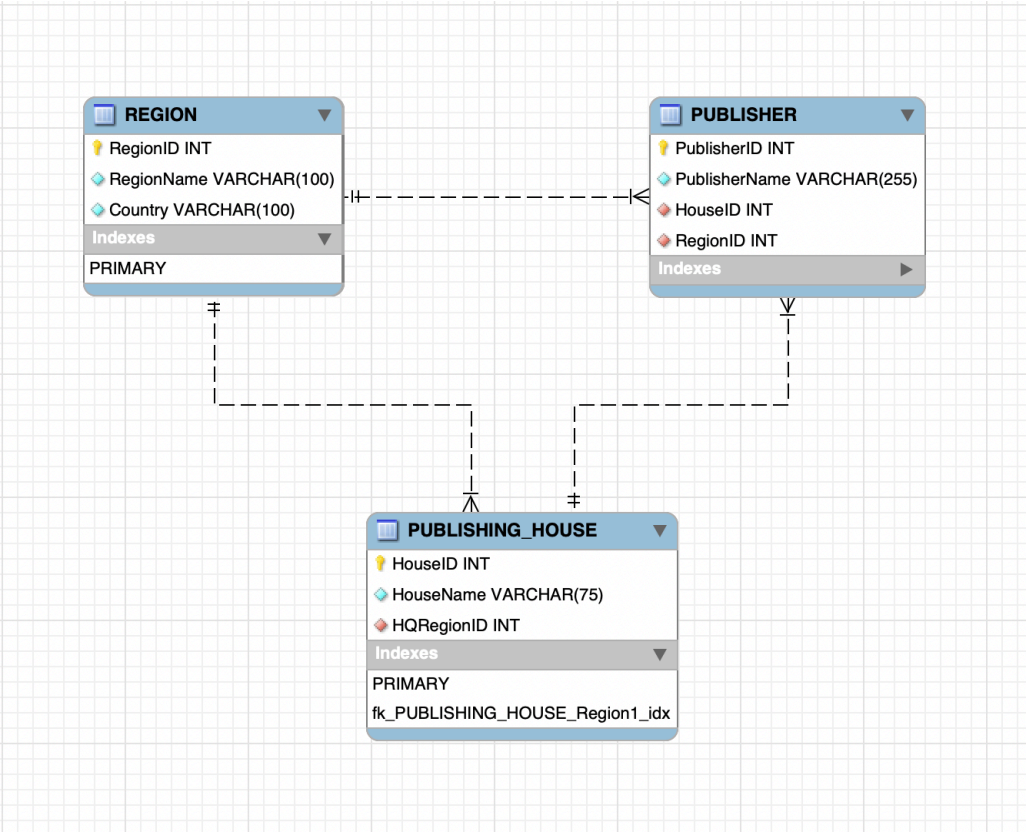
Tables:

- **REGION**(RegionID PK, RegionName, Country)
- **PUBLISHER**(PublisherID PK, PublisherName, RegionID FK)



2. A **PublishingHouse** owns multiple **Publishers** (imprint companies). Each **Publisher** belongs to exactly one **PublishingHouse**. A **PublishingHouse** may operate across multiple **Regions** via its publishers. Draw the ER diagram.

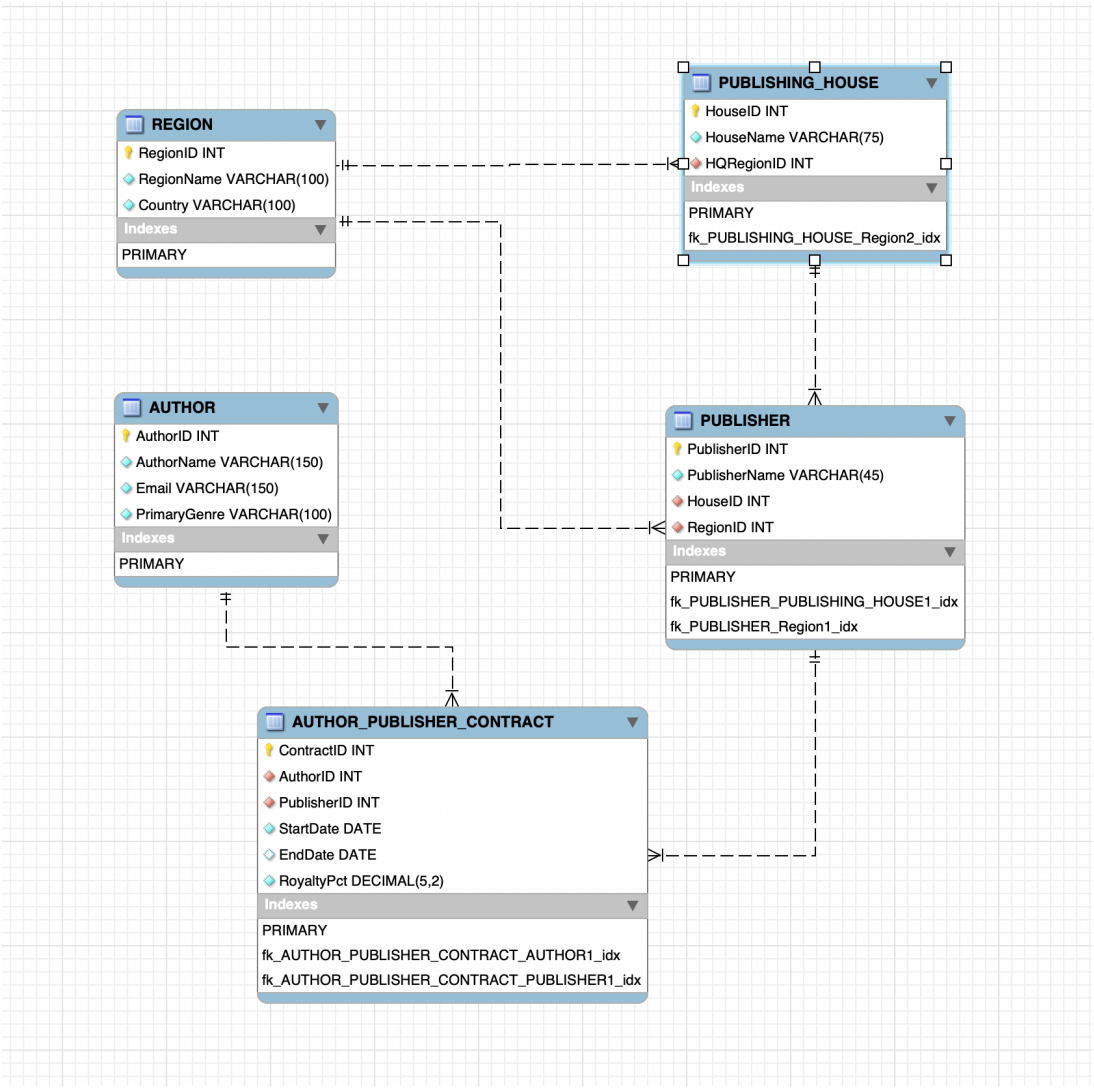
- **Tables:**
- **REGION**(RegionID PK, RegionName, Country)
- **PUBLISHING_HOUSE**(HouseID PK, HouseName, HQRegionID FK)
- **PUBLISHER**(PublisherID PK, PublisherName, HouseID FK, RegionID FK)



3.Authors can sign contracts with multiple **Publishers**, and each **Publisher** can contract multiple **Authors**. For each contract, store start date, end date (nullable), royalty percentage. Draw ER with associative entity.

Tables:

- **AUTHOR**(AuthorID PK, AuthorName, Email, PrimaryGenre)
- **PUBLISHER**(PublisherID PK, PublisherName, HouseID FK, RegionID FK)
- **AUTHOR_PUBLISHER_CONTRACT**(ContractID PK, AuthorID FK, PublisherID FK, StartDate, EndDate, RoyaltyPct)

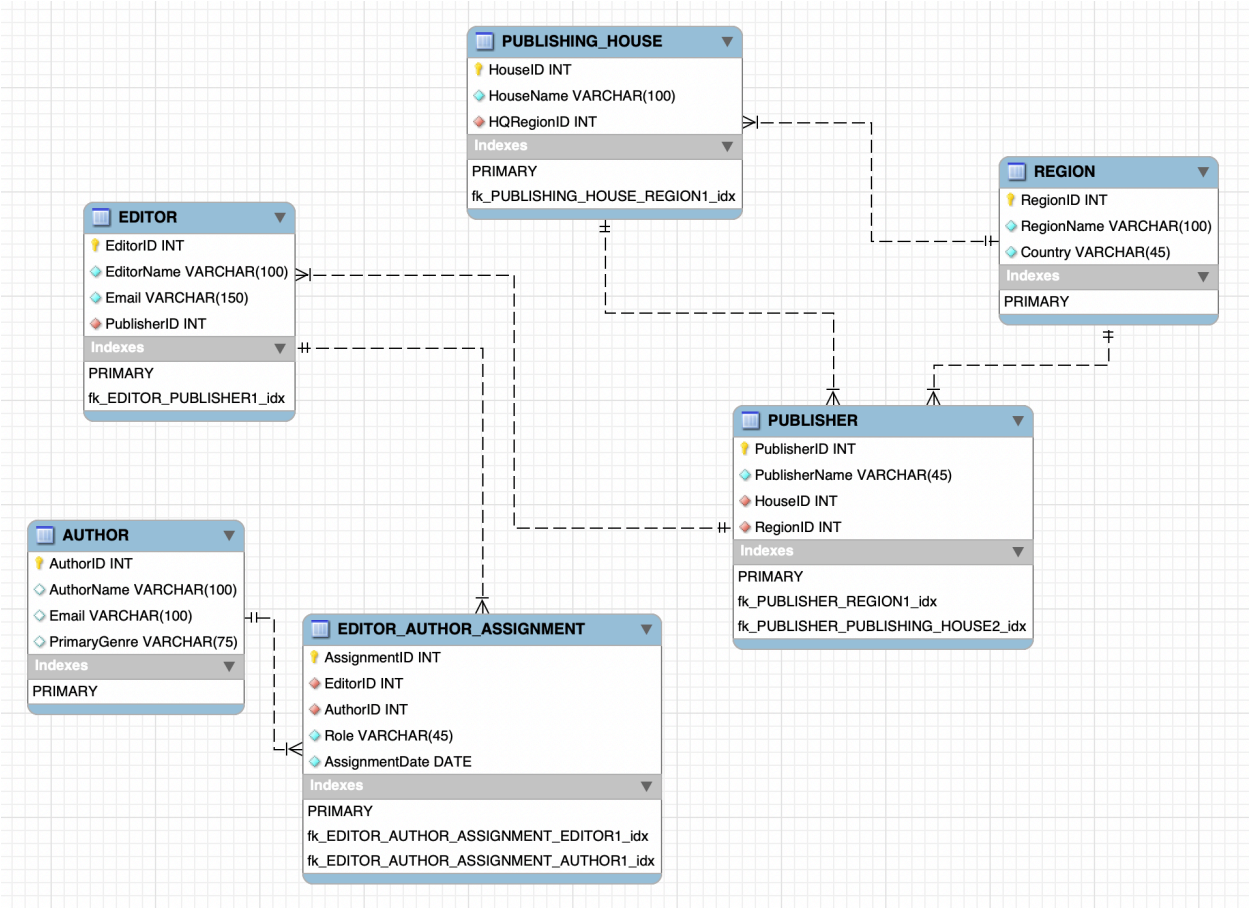


Note – REGION and Publishing House have been added in the ER diagram to show the foreign key dependencies.

4.Each **Editor** works for exactly one **Publisher**. An **Editor** can be assigned to multiple **Authors** (to manage their publications for that publisher), and an **Author** may have different editors at different publishers. Each assignment has Role (Lead/Assistant) and AssignmentDate. Draw ER with constraints.

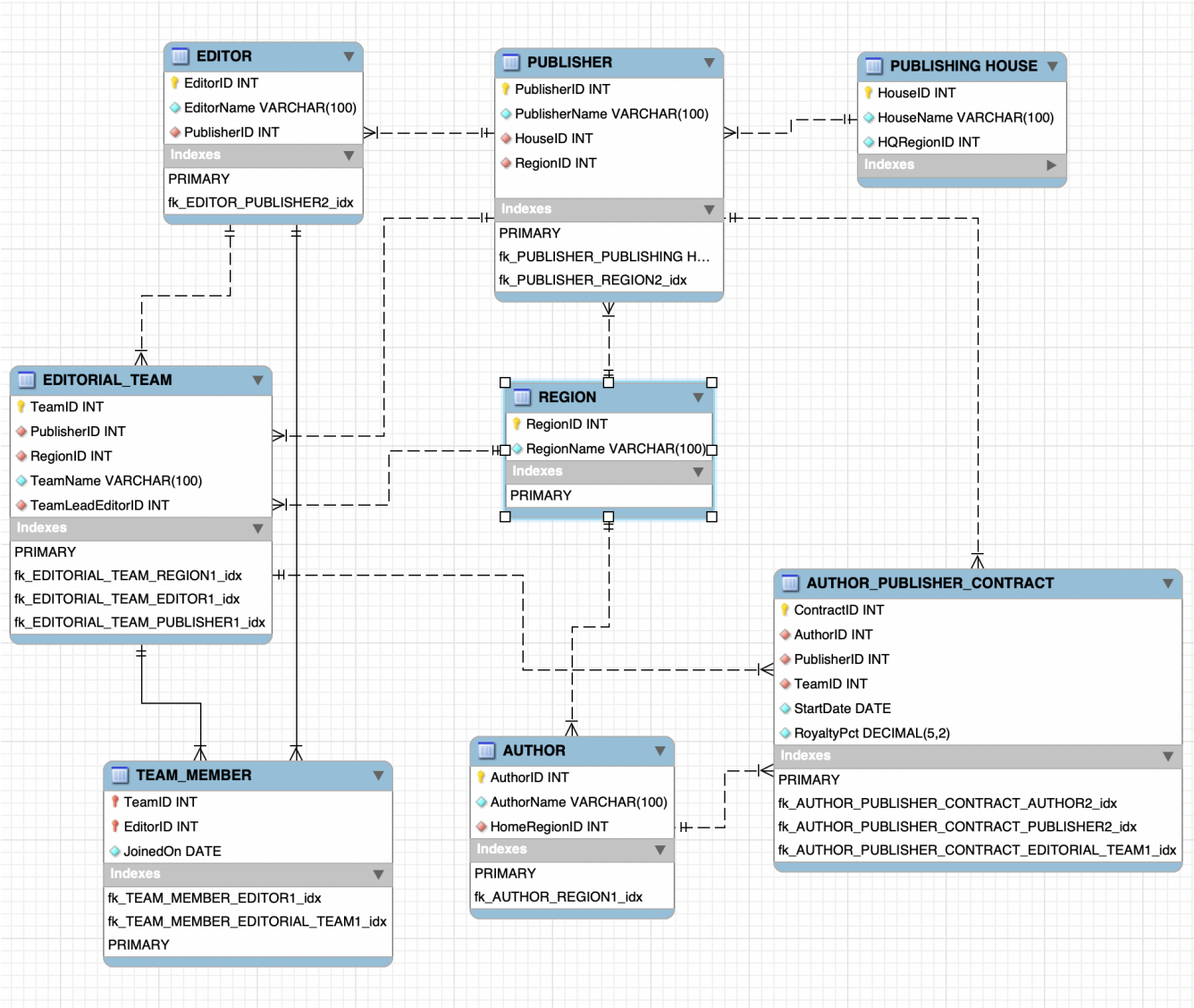
Tables:

- **EDITOR**(EditorID PK, EditorName, Email, PublisherID FK)
- **AUTHOR**(AuthorID PK, AuthorName, Email, PrimaryGenre)
- **EDITOR_AUTHOR_ASSIGNMENT**(AssignmentID PK, EditorID FK, AuthorID FK, Role, AssignmentDate)



Note – Region , Publisher and Publishing_House have been added in the ER diagram to show the foreign key dependencies.

5. A **Publisher** forms **EditorialTeams** per region for localization. A team belongs to one publisher and one region. Editors can be members of multiple teams; each team has exactly one TeamLead (who is also an editor). Authors are assigned to a team (not directly to an editor) for that contract. Draw ER diagram



NOTE- Publishing_House has been added in ER diagram to show the foreign key dependency for Publisher.HouseID.